

Year 7 · Chapter 1 · Integers: answers (teacher copy)

1.01 Integers and opposites (booklet p. 3)

Examples (model these live)

E1 Write an integer for 5 m below sea level.

below 0: negative $\cdot = -5$

E2 What is the opposite of -6 ?

-6 is 6 steps left of 0 $\cdot$ opposite: 6

E3 $-(-7)$

= the opposite of $-7 \cdot = 7$

Questions (54 on today's skill, 3 from earlier lessons)

- | | | | |
|---------------------------------------|---|---|---|
| 1 -36 | 19 = the opposite of $-17 \cdot = 17$ | 33 13 and -13 | 46 $-(-14) = 14 \cdot$ the opposite of $14 \cdot = -14$ |
| 2 20 | 20 distances: 14 and 6 $\cdot$ further: -14 | 34 $= 30 - 25 \cdot = 5$ Year 6 | 47 No. $-(-4)$ means the opposite of -4 , which is 4. |
| 3 -6 | 21 37 and -37 | 35 16 and -16 | 48 $-(-10) = 10 \cdot$ the opposite of $10 \cdot = -10$ |
| 4 10 | 22 8 and -8 | 36 = the opposite of $16 \cdot = -16$ | 49 $-6, -5, 5, 6$ |
| 5 -27 | 23 = the opposite of $15 \cdot = -15$ | 37 half of 44 is $22 \cdot$ 22 and -22 | 50 half of 38 is $19 \cdot$ 19 and -19 |
| 6 -1 | 24 distances: 3 and 10 $\cdot$ further: 10 | 38 half of 34 is $17 \cdot$ 17 and -17 | 51 15 and -15 |
| 7 -18 | 25 7 and -7 | 39 $-(-13) = 13 \cdot$ the opposite of $13 \cdot = -13$ | 52 $-(-11) = 11 \cdot$ the opposite of $11 \cdot = -11$ |
| 8 -25 | 26 12 and -12 | 40 No. The opposite of -3 is 3, which is positive. | 53 3 (the number is -3) |
| 9 1 | 27 distances: 9 and 10 $\cdot$ further: 10 | 41 half of 58 is $29 \cdot$ 29 and -29 | 54 half of 36 is $18 \cdot$ 18 and -18 |
| 10 -4 | 28 40 and -40 | 42 $-(-15) = 15 \cdot$ the opposite of $15 \cdot = -15$ | 55 7 671 Year 6 |
| 11 6 | 29 11 and -11 | 43 0. It is 0 steps from 0, so it is its own opposite. | 56 16 and -16 |
| 12 -10 | 30 = the opposite of $-18 \cdot = 18$ | 44 half of 52 is $26 \cdot$ 26 and -26 | 57 $-(-8) = 8 \cdot$ the opposite of $8 \cdot = -8$ |
| 13 = the opposite of $-12 \cdot = 12$ | 31 half of 28 is $14 \cdot$ 14 and -14 | 45 61 Year 6 | |
| 14 -19 | 32 distances: 9 and 2 $\cdot$ further: -9 | | |

1.02 The number line (booklet p. 5)

Examples (model these live)

E1 How far is it from -3 to 5?
3 to 0, then 5 from 0 $\cdot 3 + 5 = 8$

E2 Start at 4. Move 9 left. Where do you land?
4 left to 0, then 5 more $\cdot = -5$

E3 Which integer is halfway between -8 and 2?
10 apart, half is 5 $\cdot$ 5 right of -8 : -3

Questions (47 on today's skill, 9 from earlier lessons)

- | | | | |
|---|--|---|---|
| 1 2 | 18 $-5, -4$ | 30 6 1.01 | 43 each is 7 from $-7 \cdot -14$ and 0 |
| 2 -2 | 19 9 to 0, then 16 from 0 $\cdot 9 + 16 = 25$ | 31 -6 | 44 $1 \rightarrow 6 \cdot 6 \rightarrow -9$ |
| 3 12 | 20 16 apart, half is $8 \cdot 8$ right of -10 : -2 | 32 24 to 0, then 19 from 0 $\cdot 24 + 19 = 43$ | 45 distances: 5 and 14 $\cdot$ further: 14 1.01 |
| 4 $-5, -4, -3$ | 21 5, 10 | 33 each is 6 from $-9 \cdot -15$ and -3 | 46 -3 to $6 \cdot 3 + 1 + 6 = 10$ |
| 5 5 | 22 2 up to 0, then 7 more $\cdot = 9$ floors | 34 15 and -15 1.01 | 47 $5 \rightarrow 12 \cdot 12 \rightarrow 3$ |
| 6 7 | 23 -4°C Year 6 | 35 6 apart, half is $3 \cdot 3$ right of -12 : -9 | 48 -5 to $7 \cdot 5 + 1 + 7 = 13$ |
| 7 12 | 24 -19 | 36 10, 20 | 49 20 up to 0, then $1 \cdot = 21$ m |
| 8 $-4, -3, -2$ | 25 11 to 0, then 19 from 0 $\cdot 11 + 19 = 30$ | 37 200 Year 6 | 50 $\frac{1}{8}$ of 80 = $10 \cdot 4 \times 10 = 40$ Year 6 |
| 9 9 | 26 16 apart, half is $8 \cdot 8$ right of -15 : -7 | 38 each is 6 from $-3 \cdot -9$ and 3 | 51 each is 3 from $-3 \cdot -6$ and 0 |
| 10 13 | 27 16 apart, half is $8 \cdot 8$ right of -11 : -3 | 39 $2 \rightarrow 8 \cdot 8 \rightarrow -3$ | 52 -4 |
| 11 12 | 28 0, 5 | 40 -11 to $2 \cdot 11 + 1 + 2 = 14$ | 53 $-2 \rightarrow 3 \cdot 3 \rightarrow -12$ |
| 12 $-6, -5, -4, -3$ | 29 3 up to 0, then 2 more $\cdot = 5$ floors | 41 -16 1.01 | 54 -18 1.01 |
| 13 3 | | 42 count from -15 up to $-6 \cdot = 9$ m | 55 -4 to $5 \cdot 4 + 1 + 5 = 10$ |
| 14 13 to 0, then 16 from 0 $\cdot 13 + 16 = 29$ | | | 56 count from -10 up to $-1 \cdot = 9$ m |
| 15 19 1.01 | | | |
| 16 -13 | | | |
| 17 17 | | | |

1.03 Comparing and ordering integers (booklet p. 7)

Examples (model these live)

E1 Write $<$ or $>$: $-9 \square -4$
 -9 is left of $-4 \cdot -9 < -4$

E2 Order from smallest to largest: 3, $-5, 0, -1$
smallest: $-5 \cdot -5, -1, 0, 3$

E3 Which is colder: -12°C or -8°C ?
 -12 is lower than $-8 \cdot$ colder: -12°C

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|--|---|--|---|
| 1 $-10 < -6$ | 19 smallest: $-11 \cdot -11, -4, 0, 5, 10$ | 33 lowest: -10°C · coldest Cooma, warmest Oberon | 46 True. -6 is further right than -10 . |
| 2 3 | 20 largest: $6 \cdot 6, -6, -18, -28$ | 34 $\frac{1}{8}$ of $80 = 10 \cdot 5 \times 10 = 50$
Year 6 | 47 18 to 0, then 37 from 0 · $18 + 37 = 55$ ^{1.02} |
| 3 -18 | 21 0, 10 ^{1.02} | 35 largest: $5 \cdot 5, -7, -15, -27$ | 48 False. -8 is further right, so it is larger. |
| 4 -10°C | 22 $-13 > -44$ | 36 38 ^{1.01} | 49 -14 and 1 |
| 5 $-10 < -7$ | 23 $-8, -7$ | 37 $-19 > -31$ | 50 9 269 Year 6 |
| 6 1 | 24 7 Year 6 | 38 lowest: -7°C · coldest Jindabyne, warmest Cooma | 51 $-15^{\circ}\text{C}, -11^{\circ}\text{C}, -4^{\circ}\text{C}, 0^{\circ}\text{C}, 2^{\circ}\text{C}$ |
| 7 -6 | 25 smallest: $-12 \cdot -12, -3, 0, 6, 8$ | 39 True. -9 is further right than -12 . | 52 -3 ^{1.01} |
| 8 -10°C | 26 largest: $13 \cdot 13, -4, -15, -23$ | 40 -18 ^{1.02} | 53 lowest: -6°C · coldest Oberon, warmest Thredbo |
| 9 1 ^{1.02} | 27 largest: $9 \cdot 9, -2, -10, -27$ | 41 -14 and 4 | 54 $-3, -2, -1$ |
| 10 -9 | 28 $-47 < -20$ | 42 $= 38 - 2 \cdot = 36$ Year 6 | 55 14 apart, half is $7 \cdot 7$ right of -13 : -6 ^{1.02} |
| 11 -15 | 29 3 ^{1.01} | 43 No. -3 is further right on the number line, so $-3 > -8$. | 56 True. -2 is further right than -10 . |
| 12 -15°C | 30 $-7, -6$ | 44 lowest: -10°C · coldest Thredbo, warmest Oberon | 57 793 Year 6 |
| 13 71 Year 6 | 31 2 up to 0, then 9 more · = 11 floors ^{1.02} | 45 6 ^{1.01} | |
| 14 $4 > -10$ | 32 smallest: $-11 \cdot -11, -3, 0, 6, 9$ | | |
| 15 smallest: $-8 \cdot -8, -6, 0, 3, 12$ | | | |
| 16 1 | | | |
| 17 -1 ^{1.01} | | | |
| 18 -19 | | | |

1.04 Adding integers (booklet p. 9)

Examples (model these live)

- | | | |
|---|-------------------------------------|---------------------------------------|
| E1 $-3 + 5$
start at -3 , move 5 right · = 2 | E2 $4 + (-9)$
$= 4 - 9$ · = -5 | E3 $-6 + (-2)$
$= -6 - 2$ · = -8 |
|---|-------------------------------------|---------------------------------------|

Questions (43 on today's skill, 17 from earlier lessons)

- | | | | |
|-----------------------------|--|---|--|
| 1 7 | 19 -7 | 36 from -5 to -8 : 3 to the left · $\square = -3$ | 49 from -12 to -5 : 7 to the right · $\square = 7$ |
| 2 -6 | 20 -3 ^{1.03} | 37 $= 49 - 8 \cdot = 41$ Year 6 | 50 19 and -19 ^{1.01} |
| 3 8 | 21 1 | 38 $= 27 - 38 \cdot = -11$ | 51 $54 + (-65) + 22 \cdot = -11 + 22 \cdot = 11$ dollars |
| 4 -15 | 22 $= -19 - 17 \cdot = -36$ | 39 4 ^{1.01} | 52 No. Adding a negative moves left: $-8 + (-6) = -14$. |
| 5 5 | 23 $= 15 - 44 \cdot = -29$ | 40 $-9 + 13 \cdot = 4^{\circ}\text{C}$ | 53 -5°C Year 6 |
| 6 3 | 24 distances: 2 and 13 · further: 13 ^{1.01} | 41 from -10 to -6 : 4 to the right · $\square = 4$ | 54 $-1 + 5 + (-3) \cdot = 4 + (-3) \cdot =$ level 1 |
| 7 7 | 25 $-12 + 13 \cdot = 1^{\circ}\text{C}$ | 42 $47 + (-90) + 27 \cdot = -43 + 27 \cdot = -16$ dollars | 55 1 ^{1.02} |
| 8 -17 | 26 $= -3 + (-10) \cdot = -13$ | 43 1 up to 0, then 8 more · = 9 floors ^{1.02} | 56 $= 3 + (-13) + 4 \cdot = -10 + 4 \cdot = -6$ |
| 9 -1 ^{1.02} | 27 30 000 Year 6 | 44 No. Adding a negative moves left: $-3 + (-7) = -10$. | 57 from -4 to -2 : 2 to the right · $\square = 2$ |
| 10 0 | 28 $= -27 - 18 \cdot = -45$ | 45 $1/10$ Year 6 | 58 smallest: $-9 \cdot -9, -2, 0, 1, 8$ ^{1.03} |
| 11 6 | 29 $= 19 - 43 \cdot = -24$ | 46 $-100 + 100 + (-70) \cdot = 0 + (-70) \cdot = -70$ m | 59 $= 6 + (-10) \cdot = -4$ |
| 12 2 | 30 $= 9 - 32 \cdot = -23$ | 47 $= -6 + (-16) + 2 \cdot = -22 + 2 \cdot = -20$ | 60 110 Year 6 |
| 13 -3 | 31 $-10 + 8 \cdot = -2^{\circ}\text{C}$ | 48 -11°C ^{1.03} | |
| 14 -7 | 32 14 ^{1.02} | | |
| 15 -11 | 33 $= 0 + (-11) \cdot = -11$ | | |
| 16 10 Year 6 | 34 $-21 < -20$ ^{1.03} | | |
| 17 -1 | 35 $= -27 - 24 \cdot = -51$ | | |
| 18 $= -38 - 19 \cdot = -57$ | | | |

1.05 Subtracting integers (booklet p. 11)

Examples (model these live)

- | | | |
|--|----------------------------------|------------------------------------|
| E1 $2 - 7$
start at 2, move 7 left · = -5 | E2 $5 - (-3)$
$= 5 + 3$ · = 8 | E3 $-4 - (-9)$
$= -4 + 9$ · = 5 |
|--|----------------------------------|------------------------------------|

Questions (42 on today's skill, 17 from earlier lessons)

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|---------------------|----------------------------|--|--|
| 1 -6 | 10 -4 | 19 11 | 27 $= -37 + 49 \cdot = 12$ |
| 2 9 | 11 5 | 20 3 ^{1.03} | 28 6 apart, half is $3 \cdot 3$ right of -8 : -5 ^{1.02} |
| 3 -11 | 12 -11 | 21 -6 | 29 $= -14 + 6 \cdot = -8$ |
| 4 -2 | 13 8 | 22 -2 | 30 $4 - (-8) \cdot = 4 + 8 \cdot = 12^{\circ}\text{C}$ |
| 5 -4 | 14 -16 | 23 $= -11 + 21 \cdot = 10$ | 31 $12 - (-10) \cdot = 12 + 10 \cdot = 22$ |
| 6 11 | 15 -3 | 24 $13 - (-10) \cdot = 13 + 10 \cdot = 23^{\circ}\text{C}$ | 32 $= 22 - 6 \cdot = 16$ Year 6 |
| 7 -12 | 16 $1/10$ Year 6 | 25 distances: 4 and 10 · further: 10 ^{1.01} | 33 $= -22 + 6 \cdot = -16$ |
| 8 7 | 17 -2 | 26 $28 - (-26) \cdot = 28 + 26 \cdot = 54$ | 34 7 ^{1.01} |
| 9 5 ^{1.04} | 18 $= -17 + 7 \cdot = -10$ | | 35 $= -13 + 6 \cdot = -7$ |

$$36 \quad \square + 4 = 0 \cdot \square = -4$$

$$37 \quad 10 \text{ 1.04}$$

$$38 \quad 15 - (-2) \cdot = 15 + 2 \cdot = 17^\circ\text{C}$$

$$39 \quad -28 < -11 \text{ 1.03}$$

$$40 \quad \square + 2 = 1 \cdot \square = -1$$

$$41 \quad = 2 + 5 - 5 \cdot = 7 - 5 \cdot = 2$$

$$42 \quad 7 \text{ 624 Year 6}$$

$$43 \quad 760 - (-180) \cdot = 760 + 180 \cdot = 940 \text{ m}$$

$$44 \quad 11 \text{ 1.02}$$

$$45 \quad \text{No. Subtracting a negative moves right: } 6 + 4 = 10.$$

$$46 \quad \square + 3 = 5 \cdot \square = 2$$

$$47 \quad 5 \text{ and } -5 \text{ 1.01}$$

$$48 \quad = -2 + 6 - 10 \cdot = 4 - 10 \cdot = -6$$

$$49 \quad = -2 + (-7) \cdot = -9 \text{ 1.04}$$

$$50 \quad = -4 + 4 - 5 \cdot = 0 - 5 \cdot = -5$$

$$51 \quad 320 - (-190) \cdot = 320 + 190 \cdot = 510 \text{ m}$$

$$52 \quad 11 \text{ Year 6}$$

$$53 \quad \text{No. Subtracting a negative moves right: } 5 + 7 = 12.$$

$$54 \quad 15 \text{ 1.01}$$

$$55 \quad \square + 8 = 9 \cdot \square = 1$$

$$56 \quad = -35 + 23 \cdot = -12$$

$$57 \quad -19^\circ\text{C 1.03}$$

$$58 \quad = -2 + 4 - 14 \cdot = 2 - 14 \cdot = -12$$

$$59 \quad 1 \text{ up to } 0, \text{ then } 6 \text{ more } \cdot = 7 \text{ floors 1.02}$$

1.06 Multiplying integers (booklet p. 13)

Examples (model these live)

E1 -4×3
different signs: $- \cdot = -12$

E2 $-5 \times (-6)$
same signs: $+ \cdot = 30$

E3 $-2 \times 3 \times (-4)$
 $= -6 \times (-4) \cdot = 24$

Questions (43 on today's skill, 17 from earlier lessons)

1 -15	17 -64	32 $9/10 \text{ Year 6}$	47 $30 \div (-5) \cdot \square = -6$
2 6	18 same signs: $+ \cdot = 45$	33 $3 \times (-5) \cdot = -15^\circ\text{C}$	48 $19 \text{ and } -19 \text{ 1.01}$
3 positive	19 40	34 2 1.02	49 try factor pairs of $-48 \cdot 8$ and -6
4 $(-4) + (-4) + (-4) + (-4) + (-4) \cdot = -20$	20 12 1.05	35 same signs: $+ \cdot = 84$	50 $= -3 + (-8) \cdot = -11 \text{ 1.04}$
5 -12	21 positive	36 $-49 \div (-7) \cdot \square = 7$	51 try factor pairs of $-40 \cdot 8$ and -5
6 81	22 same signs: $+ \cdot = 126$	37 -5 1.04	52 $7 \text{ negatives: odd } \cdot = -1$
7 positive	23 $= -1 \times 5 \cdot = -5$	38 $= -2 \times 2 \cdot = -4$	53 100 Year 6
8 $(-3) + (-3) + (-3) \cdot = -9$	24 distances: $4 \text{ and } 11 \cdot$ further: 11 1.01	39 $26 - (-10) \cdot = 26 + 10 \cdot = 36 \text{ 1.05}$	54 No. $(-8)^2 = -8 \times (-8) = 64$: same signs, positive.
9 2 1.04	25 $= -12 \times (-12) \cdot = 144$	40 $= -2 \times (-2) \cdot = 4$	55 7 1.02
10 -9	26 $5 \times (-3) \cdot = -15^\circ\text{C}$	41 $16 \div 2 \cdot \square = 8$	56 $63 \div (-7) \cdot \square = -9$
11 48	27 largest: $3 \cdot 3, -3, -10, -21 \text{ 1.03}$	42 try factor pairs of $-56 \cdot 7$ and -8	57 $6 \times (-6) \cdot = -36^\circ\text{C}$
12 negative	28 same signs: $+ \cdot = 72$	43 -8°C Year 6	58 -2 1.05
13 72	29 $= 12 \times (-5) \cdot = -60$	44 $6 \text{ negatives: even } \cdot = 1$	59 try factor pairs of $-8 \cdot 4$ and -2
14 positive	30 $= -8 \times (-3) \cdot = 24$	45 -12 1.03	60 $-7, -6, -5 \text{ 1.03}$
15 $(-2) + (-2) \cdot = -4$	31 $= -3 \times (-3) \cdot = 9$	46 No. $(-8)^2 = -8 \times (-8) = 64$: same signs, positive.	
16 $= 28 - 3 \cdot = 25 \text{ Year 6}$			

1.07 Dividing integers (booklet p. 15)

Examples (model these live)

E1 $-20 \div 4$
different signs: $- \cdot = -5$

E2 $-36 \div (-9)$
same signs: $+ \cdot = 4$

E3 Find the average: $-3, 5, -8, -6$
total: $-12 \cdot = -12 \div 4 \cdot = -3$

Questions (42 on today's skill, 17 from earlier lessons)

1 -8	17 -7	32 $= 22 - 4 \cdot = 18 \text{ Year 6}$	46 $\square = 5 \times 3 \cdot \square = 15$
2 9	18 same signs: $+ \cdot = 11$	33 $= -81 \div 9 \cdot = -9$	47 largest: $3 \cdot 3, -7, -13, -23 \text{ 1.03}$
3 -9	19 4	34 -4 1.02	48 $\square = -14 \div 7 \cdot \square = -2$
4 negative	20 3 1.05	35 same signs: $+ \cdot = 11$	49 $4 \times (-5) \cdot = -20^\circ\text{C 1.06}$
5 -9	21 -2	36 $\square = 1 \times 7 \cdot \square = 7$	50 $\square = 18 \div (-9) \cdot \square = -2$
6 3	22 positive	37 negative 1.06	51 $= -4 \times (-4) \cdot = 16$
7 -2	23 different signs: $- \cdot = -12$	38 total: $0 \cdot = 0 \div 4 \cdot = 0^\circ\text{C}$	52 8 Year 6
8 negative	24 total: $-5 \cdot = -5 \div 5 \cdot = -1^\circ\text{C}$	39 $21 - (-12) \cdot = 21 + 12 \cdot = 33 \text{ 1.05}$	53 No. Same signs give a positive: $-18 \div (-3) = 6$.
9 49 1.06	25 distances: $11 \text{ and } 14 \cdot$ further: 14 1.01	40 $\square = -12 \times 3 \cdot \square = -36$	54 6 1.01
10 -4	26 $-42 \div 7 \cdot = -6 \text{ m}$	41 $\square = -12 \div 4 \cdot \square = -3$	55 $\square = 8 \times (-1) \cdot \square = -8$
11 2	27 $= -12 \div (-6) \cdot = 2$	42 -5°C Year 6	56 $= -81 \div (-9) \cdot = 9$
12 -4	28 $= 22 - 15 \cdot = 7 \text{ 1.04}$	43 $= -6 \times (-2) \cdot = 12$	57 0 1.05
13 7	29 same signs: $+ \cdot = 11$	44 -1 1.04	58 $\square = -6 \div 2 \cdot \square = -3$
14 -7	30 total: $-4 \cdot = -4 \div 4 \cdot = -1^\circ\text{C}$	45 No. Same signs give a positive: $-49 \div (-7) = 7$.	59 $= 10 + (-7) \cdot = 3 \text{ 1.04}$
15 positive	31 $-35 \div 7 \cdot = -5 \text{ m}$		
16 $3/10 \text{ Year 6}$			

1.08 Order of operations with integers (booklet p. 17)

Examples (model these live)

E1 $-3 + 4 \times (-2)$
 $= -3 + (-8) \cdot = -11$

E2 $(-6 + 2) \times 5$
 $= -4 \times 5 \cdot = -20$

E3 $-20 \div (3 - 8)$
 $= -20 \div (-5) \cdot = 4$

Questions (40 on today's skill, 17 from earlier lessons)

- | | | | |
|--|---|---|--|
| 1 $= -4 + (-3) \cdot = -7$ | 17 $8^{1.07}$ | 32 $= 1 + (-10) \cdot = -9$ | 46 $= [-8 + (-12)] \div 4 \cdot = -20 \div 4 \cdot = -5$ |
| 2 $= -12 - 5 \cdot = -17$ | 18 $= 1 + 3 \cdot = 4$ | 33 $= 1 \times 7 \cdot = 7$ | 47 $8 \times (-5) \cdot = -40^\circ\text{C}^{1.06}$ |
| 3 $= 5 \times (-4) \cdot = -20$ | 19 $= 6 + (-12) \cdot = -6$ | 34 negative ^{1.06} | 48 $= [-3 + (-15)] \div 3 \cdot = -18 \div 3 \cdot = -6$ |
| 4 $= 9 + 15 \cdot = 24$ | 20 $= -7 - 13 \cdot = -20$ | 35 $= 2 - (-2) \cdot = 4$ | 49 $(-9 + 5) \times 2 \cdot = -4 \times 2 = -8$ |
| 5 $= 30 - 10 \cdot = 20$ | 21 distances: 15 and 5 | 36 $-96 \div 8 \cdot = -12 \text{ m}^{1.07}$ | 50 $7/10$ ^{Year 6} |
| 6 $= 2 \times 4 \cdot = 8$ | further: -15 ^{1.01} | 37 $= 16 - 19 \cdot = -3$ | 51 No. $\times$ comes first: 7 - (-15) = 22. |
| 7 $= 7 + 8 \cdot = 15$ | 22 $= 36 - 30 \cdot = 6$ | 38 $= 11 \times (-7) \cdot = -77$ | 52 $4^{1.04}$ |
| 8 $= -18 - 10 \cdot = -28$ | 23 $= 5 \div (-5) \cdot = -1$ | 39 $= [-7 + 4] \div 3 \cdot = -3 \div 3 \cdot = -1$ | 53 $= (-15) \times 5 \cdot = -75$ |
| 9 $36^{1.06}$ | 24 $9 - (-5) \cdot = 9 + 5 \cdot = 14^\circ\text{C}^{1.05}$ | 40 25 students ^{Year 6} | 54 $= -15 \div (-3) \cdot = 5$ |
| 10 $= 10 - 12 \cdot = -2$ | 25 $= 4 + 5 \cdot = 9$ | 41 $(-5 + 3) \times 3 \cdot = -2 \times 3 = -6$ | 55 positive ^{1.07} |
| 11 $= -9 \times 2 \cdot = -18$ | 26 $= -11 - 11 \cdot = -22$ | 42 $-9^{1.05}$ | 56 $= [-5 + (-15)] \div 5 \cdot = -20 \div 5 \cdot = -4$ |
| 12 $= 8 + 24 \cdot = 32$ | 27 $= -8 - 12 \cdot = -20$ | 43 No. $\times$ comes first: 7 - (-8) = 15. | 57 $= -35 + 14 \cdot = -21^{1.05}$ |
| 13 $= 24 - 3 \cdot = 21$ ^{Year 6} | 28 $= 9 - 13 \cdot = -4$ | 44 $= (-4) \times 12 \cdot = -48$ | |
| 14 $= -4 + (-25) \cdot = -29$ | 29 90 000 ^{Year 6} | 45 largest: 1 $\cdot$ 1, -5, -16, -26 ^{1.03} | |
| 15 $= -15 - 5 \cdot = -20$ | 30 $= -6 \div (-3) \cdot = 2$ | | |
| 16 $= 3 \times 1 \cdot = 3$ | 31 $-5^{1.02}$ | | |

1.09 Integers on a calculator (booklet p. 19)

Examples (model these live)

E1 Which keys do you press for -8×5 ?
 $(-)\ 8 \times 5 = \cdot = -40$

E2 Use a calculator: $-348 + 129$
 $(-)\ 348 + 129 = \cdot = -219$

E3 Estimate, then use a calculator:
 -394×21
 $\approx -400 \times 20 = -8\ 000 \cdot = -8\ 274$

Questions (39 on today's skill, 17 from earlier lessons)

- | | | | |
|--|--|--|--|
| 1 -332 | 16 $-1\ 748$ | 30 $-1\ 254 \div 38 = -33 \cdot = -315$ | 43 $((-) 4) x^2 \cdot = 16$ |
| 2 -209 | 17 $7^{1.07}$ | 31 $2^{1.02}$ | 44 largest: 13 $\cdot$ 13, -9, -14, -26 ^{1.03} |
| 3 $(-) 7 + 2 =$ | 18 $(-) 6 + 4 =$ | 32 $\approx -50 \times 40 = -2\ 000 \cdot = -1\ 980$ | 45 bracket: 112 $\cdot = -8$ |
| 4 33 | 19 $\approx -50 \times 30 = -1\ 500 \cdot = -1\ 421$ | 33 $((-) 8) x^2 \cdot = 64$ | 46 $= 2 \div (-1) \cdot = -2^{1.08}$ |
| 5 120 | 20 $((-) 23) x^2 = \cdot = 529$ | 34 $= -4 \times (-2) \cdot = 8^{1.08}$ | 47 bracket: -276 $\cdot = 23$ |
| 6 -880 | 21 distances: 13 and 9 | 35 $((-) 12) x^2 = \cdot = 144$ | 48 $13 \times (-34) \cdot = -442^\circ\text{C}$ |
| 7 $(-) 6 \times 4 =$ | further: -13 ^{1.01} | 36 $-30 \div 3 \cdot = -10 \text{ m}^{1.07}$ | 49 64 ^{Year 6} |
| 8 22 | 22 bracket: -228 $\cdot = 2\ 508$ | 37 $((-) 7) x^2 \cdot = 49$ | 50 No. $\approx -300 \times 70 = -21\ 000$; that estimate is 10 times too big. |
| 9 $= 2 - 7 \cdot = -5^{1.08}$ | 23 $-1\ 295 \div 37 = -35 \cdot = -331$ | 38 bracket: -342 $\cdot = 18$ | 51 $-1^{1.04}$ |
| 10 -810 | 24 $= 5 \times 6 \cdot = 30^{1.06}$ | 39 $\frac{1}{8}$ of 80 = $10 \cdot 6 \times 10 = 60$ ^{Year 6} | 52 $((-) 4) x^2 \cdot = 16$ |
| 11 $(-) 5 - 8 =$ | 25 $\approx -50 \times 20 = -1\ 000 \cdot = -1\ 144$ | 40 $14 \times (-15) \cdot = -210^\circ\text{C}$ | 53 $-400 \div 16 = -25 \cdot = -313$ |
| 12 33 | 26 $((-) 20) x^2 = \cdot = 400$ | 41 positive ^{1.06} | 54 positive ^{1.07} |
| 13 11 ^{Year 6} | 27 $((-) 17) x^2 = \cdot = 289$ | 42 No. $\approx -200 \times 70 = -14\ 000$; that estimate is 10 times too big. | 55 bracket: 522 $\cdot = -29$ |
| 14 72 | 28 bracket: -71 $\cdot = 994$ | | 56 $4 \times (-2) \cdot = -8^\circ\text{C}^{1.06}$ |
| 15 $\approx -40 \times 30 = -1\ 200 \cdot = -1\ 218$ | 29 $= 39 - 5 \cdot = 34$ ^{Year 6} | | |

1.10 Integer problems (booklet p. 21)

Examples (model these live)

E1 It is -4°C . The temperature rises 9°C . What is it now?
 $-4 + 9 \cdot = 5^\circ\text{C}$

E2 A diver at -12 m goes down 7 m , then up 15 m . Where is the diver now?
 $-12 - 7 + 15 \cdot = -19 + 15 \cdot = -4 \text{ m}$

E3 The temperature falls 3°C each hour for 5 hours. What is the change?
 $5 \times (-3) \cdot = -15^\circ\text{C}$

Questions (33 on today's skill, 12 from earlier lessons)

- | | | | |
|--------------------------------------|--|---|---|
| 1 $0 - 13 \cdot = -13^\circ\text{C}$ | 12 $-5 - 8 + 20 \cdot = -13 + 20 \cdot = 7 \text{ m}$ | 22 $5 \times (-5) \cdot = -25 \text{ cm}$ | 31 $6\ 881$ ^{Year 6} |
| 2 $47 - 68 \cdot = -21$ dollars | 13 $34 - 64 \cdot = -30$ dollars | 23 $= 28 - 6 \cdot = 22$ ^{Year 6} | 32 $-51 + 4 \times 25 \cdot = -51 + 100 \cdot = 49$ dollars |
| 3 $-4 + 12 \cdot =$ level 8 | 14 $-1 + 11 \cdot =$ level 10 | 24 $(-3) + 1 + (-2) \cdot = -4$ | 33 $-5^{1.04}$ |
| 4 $3 - 4 \cdot = -1^\circ\text{C}$ | 15 $-20 - 7 + 8 \cdot = -27 + 8 \cdot = -19 \text{ m}$ | 25 $14^{1.02}$ | 34 $277 - (-110) \cdot = 277 + 110 \cdot = 387 \text{ m}$ |
| 5 $77 - 119 \cdot = -42$ dollars | 16 $9 - (-11) \cdot = 9 + 11 \cdot = 20^\circ\text{C}$ | 26 $-13 - 8 + 25 \cdot = -21 + 25 \cdot = 4 \text{ m}$ | 35 $7 \times (-4) = -28 \cdot 10 + (-28) \cdot = -18^\circ\text{C}$ |
| 6 $-4 + 5 \cdot =$ level 1 | 17 distances: 8 and 9 | 27 $5 \times (-2) = -10 \cdot 6 + (-10) \cdot = -4^\circ\text{C}$ | 36 largest: 14 $\cdot$ 14, -8, -17, -28 ^{1.03} |
| 7 $6 - 5 \cdot = 1^\circ\text{C}$ | further: 9 ^{1.01} | 28 $= 10 - 6 \cdot = 4^{1.08}$ | 37 total: -12 $\cdot = -12 \div 4 \cdot = -3^\circ\text{C}$ |
| 8 $42 - 56 \cdot = -14$ dollars | 18 $5 \times (-6) \cdot = -30 \text{ cm}$ | 29 $8 \times (-4) = -32 \cdot 12 + (-32) \cdot = -20^\circ\text{C}$ | |
| 9 $-1 + 8 \cdot =$ level 7 | 19 $(-2) + 4 + 0 \cdot = 2$ | 30 total: -16 $\cdot = -16 \div 4 \cdot = -4^\circ\text{C}$ | |
| 10 $4 - 7 \cdot = -3^\circ\text{C}$ | 20 $((-) 12) x^2 = \cdot = 144^{1.09}$ | | |
| 11 10 ^{Year 6} | 21 $4 - (-3) \cdot = 4 + 3 \cdot = 7^\circ\text{C}$ | | |

38 $-73 + 3 \times 17 \cdot = -73 + 51 \cdot =$
 -22 dollars

39 774 Year 6

40 $401 - (-229) \cdot = 401 + 229 \cdot$
 $= 630$ m

41 12 1.06

42 $8 \times (-2) = -16 \cdot 11 + (-16) \cdot$
 $= -5^{\circ}\text{C}$

43 $(-6) + 4 + 2 \cdot = 0$

44 $(-) 6 - 6 = 1.09$

45 total: $-36 \cdot = -36 \div 4 \cdot =$
 -9°C